Siddhi Sanjay More

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Education

Northeastern University, Boston, MA **May 2027**
Master of Science in Robotics with Concentration in Mechanical Engineering GPA: 3.67
Awards: Global Student Award | **Relevant Courses**: Robot Sensing and Navigation, Robotics Mechanics and Control, Assistive Robotics, Wearable Robotics, Control Systems Engineering

Mumbai University, Mumbai, India **Jul 2020 - Jul 2024**
Bachelor of Engineering in Mechanical Engineering
Honors: Robotics | **Relevant Courses**: Finite Element Analysis, Thermal Engineering, Machine Design, Logistics and Supply Chain Management, Fluid Mechanics, CAD/ CAM, Automation and AI

Technical Skills and Core Competencies

Hardware and Engineering: Autodesk AutoCAD, CATIA, ANSYS Workbench, SolidWorks, Autodesk Fusion 360, MATLAB/ Simulink, XFLR5, NX 11, Arduino, Microcontrollers
Software and Programming: Python, C, MATLAB, Git/Version Control, Linux (Ubuntu)
Robotics and Controls: ROS/ROS2, Nav2, SLAM Toolbox, MATLAB, Robo-Analyzer

Experience

Robotics Intern, *IIT Bombay and 1Stop.ai, India* **Feb 2022 – Apr 2022**

- Developed **kinematic and inverse kinematic models** for a SCARA Robot to compute precise end-effector coordinates using **MATLAB**
- Simulated **structural integrity and payload capacity** within **ANSYS Workbench** to evaluate maximum **stress and deformation** for mechanical stability
- Analyzed 3R manipulator workspace by **calculating DH parameters and transformation matrices** to optimize robotic arm reach and movement efficiency
- Navigated **motion control configurations** in **Robo-Analyzer** to visualize joint end-effector relationships for enhanced industrial precision

Projects

Sound Triangulation and Doppler - Motion Detection, *Northeastern University* **Sept 2025 – Dec 2025**

- Designed and simulated a wearable **assistive robotic system** in **MATLAB** that identifies and tracks approaching vehicles to **provide haptic feedback** for hearing impaired users
- Systemized an **FFT- based signal processing pipeline** that **achieved 95% accuracy** in classifying safety-critical sounds (horns and sirens) against ambient urban noises
- Programmed **a GCC-PHAT TDOA localization algorithm** in **Python** that successfully detected vehicle direction within **10 degrees accuracy** range within a **system latency of <100ms**

Trajectory Generation on for SLAM with ROS2 and Turtlebot3*, Northeastern University* **Jan 2025– Apr 2025**

- Engineered a **ROS2 control package** for Turtlebot3 that navigated closed-loop trajectories to automate **SLAM mapping**
- Devised a **navigation stack utilizing Nav2**, collision detection and velocity control to detect and eliminate manual mapping errors, **ensuring consistent map density**
- Validated system reliability by **calculating odometry errors** and **trajectory accuracy** across varied environments, optimizing the **control systems** for superior autonomous performance

Plastic Injection Molding Analysis using FEA & DOE, *Mumbai University, India* **Jul 2023 - Oct 2024**

- Optimized the manufacturing process for automotive airbag housings using **SolidWorks Plastics,** enhancing **dimensional accuracy and mechanical integrity** for **high-pressure safety components**
- Formulated a systematic **Design of Experiments (DOE) and ANOVA framework** to identify critical process parameters reducing sink marks by 20% and warpage by 11%
- **Authored a research paper**, "Experimentation in Injection Molding: Simulation and Parameter Optimization Using DOE and ANOVA", accepted for publication in the **AIP Conference Proceedings** (ICAPSM 2024)

Activities

ASME Student Representative, *Mumbai, India* **Mar 2022 – Aug 2022**